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1 Getting Started

1.1 Accessing the Platform

- Open a supported web browser (Chrome 84+, Firefox 103+, Edge 84+, Safari 15+)
- Navigate to the CodeClash Arena platform
- Ensure JavaScript is enabled
- The homepage loads with title "HELLO CODERS, WELCOME!"
- On the homepage, you will see: **GET STARTED** button, **About us** link, and **Learn More** link

1.2 Sign-Up Process

1.2.1 Step 1: Navigate to Sign-Up Page

- Click **"GET STARTED"** on the homepage
- Click **"Sign Up"** link at the bottom of the Login page

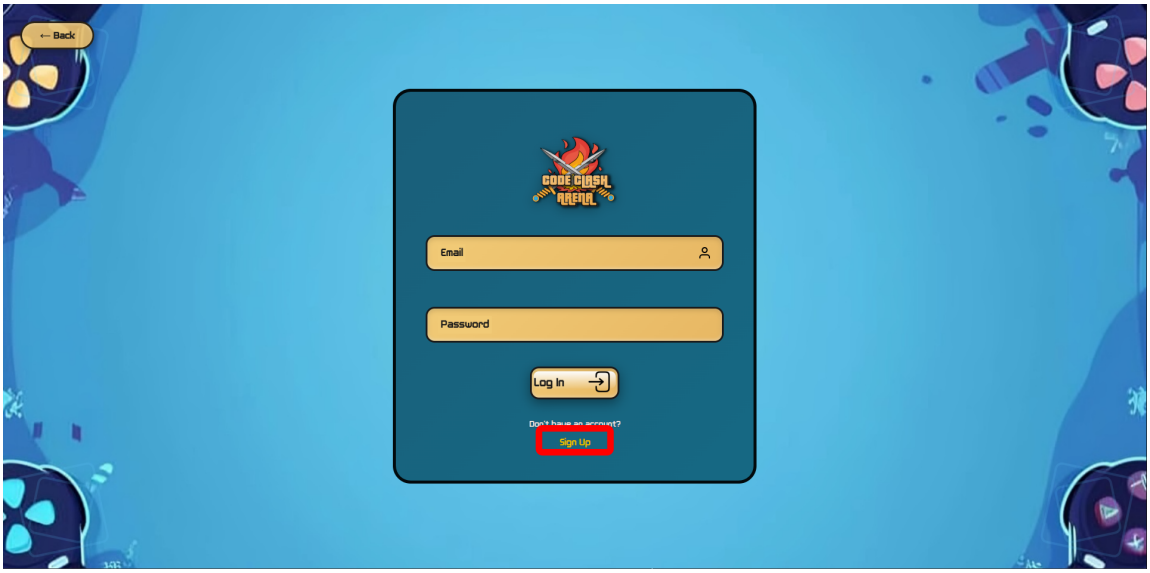


Figure 1.1: CodeClash Arena Loginpage - Locate Sign Up button

1.2.2 Step 2: Enter Registration Information

- **Email** - For verification and password recovery; must be unique
- **Codeforces Handle** - Your existing Codeforces username (create at `https://codeforces.com/register` if needed)
- **Password** - Minimum 8 characters; keep separate from Codeforces password

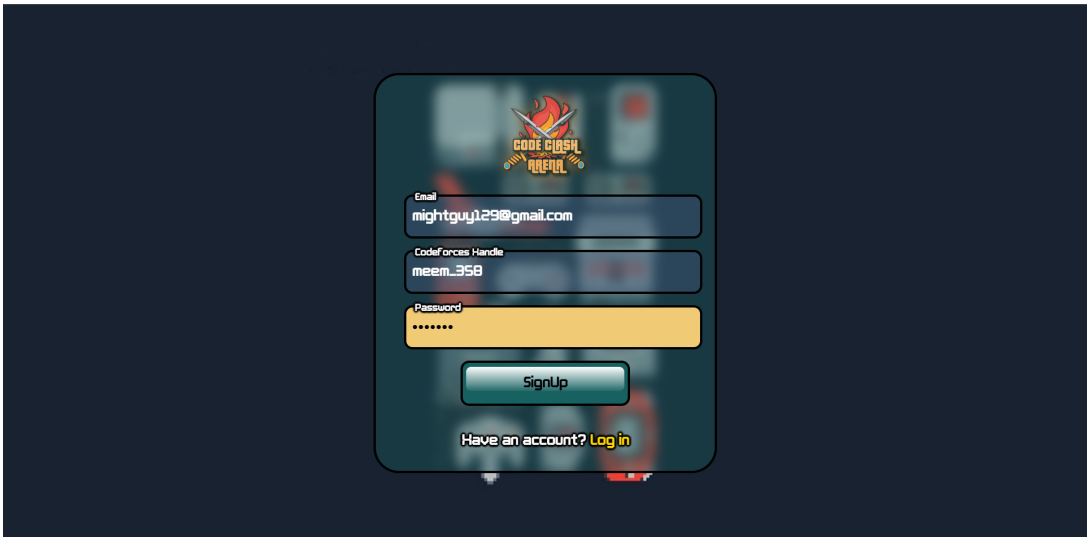


Figure 1.2: Sign-Up Form - Enter registration details

1.2.3 Step 3: Submit Registration

- Click **"SignUp"** button
- If validation fails, red field labels appear; correct errors and resubmit
- Success message: **"Signup successful! Check your email"**

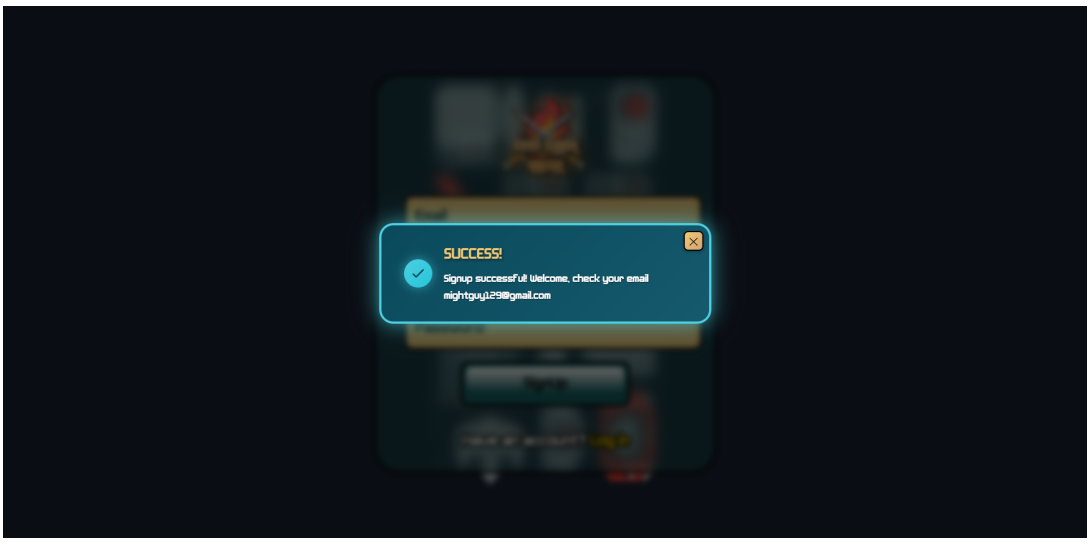


Figure 1.3: Sign-Up Form Completion - Click SignUp button

1.2.4 Step 4: Email Verification

- Check email inbox for verification message from Supabase
- Email may take 1-5 minutes to arrive; check spam folder if needed
- Click **"Verify Email Address"** link in the email
- Your account is now activated

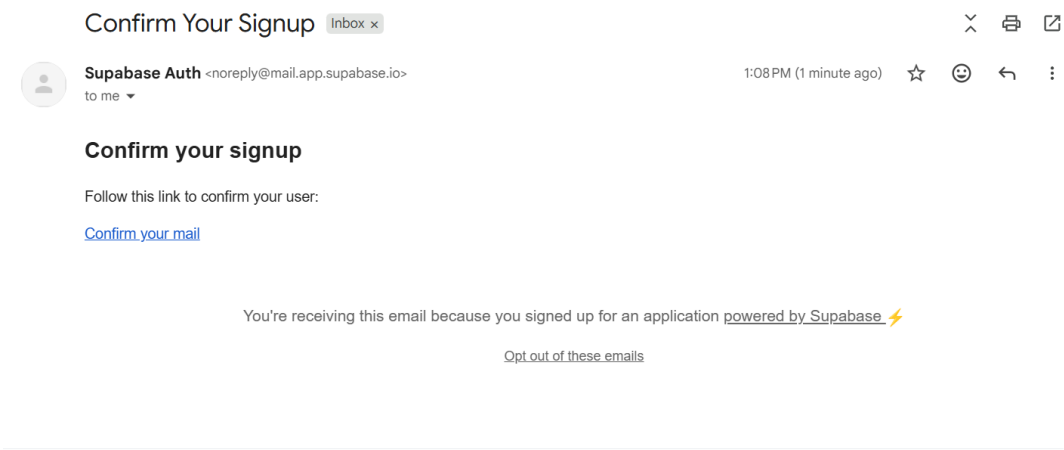


Figure 1.4: Email Verification - Click verification link

1.2.5 Step 5: Login to Complete Setup

- Navigate to Login page and use your registered email and password
- Your Codeforces handle is automatically linked from registration
- Requirements: Existing Codeforces account and verified email before login

1.3 Login Process

1.3.1 Step 1: Navigate to Login Page

- Click **"GET STARTED"** on the homepage
- You are redirected to the Login page

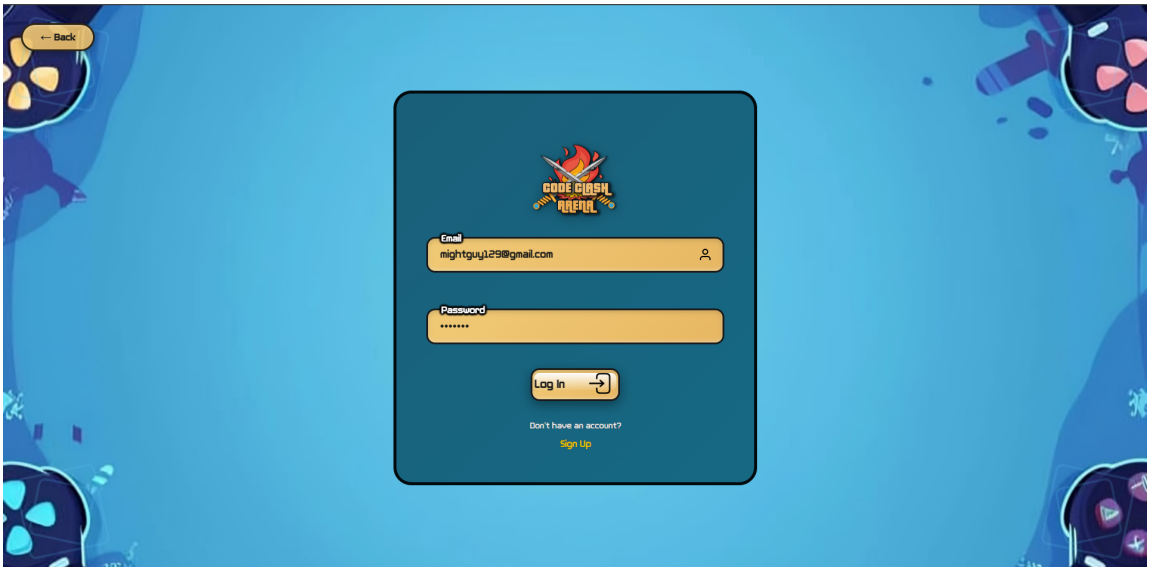


Figure 1.5: Loginpage - Navigate to Login

1.3.2 Step 2: Enter Login Credentials

- Enter your registered email address (case-insensitive)
- Enter your account password (case-sensitive)
- A visibility icon appears in the password field to show/hide password

1.3.3 Step 3: Submit and Authenticate

- Click **"Log In"** button
- System validates credentials with Supabase
- Success message: **"Sign in successful!"**

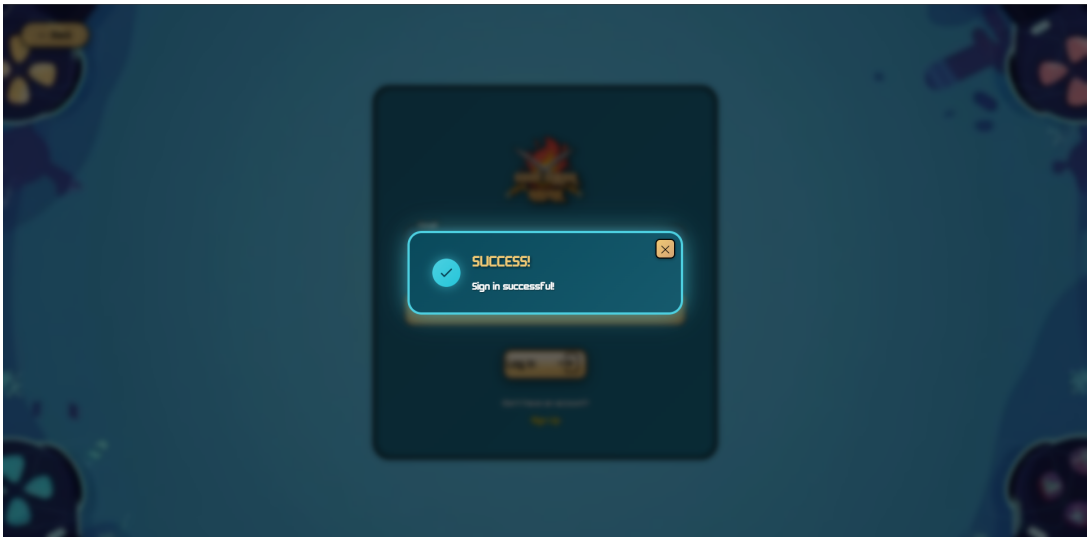


Figure 1.6: Login successful - Success alert displayed

1.3.4 Step 4: Access Dashboard

- After successful login, you are redirected to the Main Dashboard
- Your Codeforces handle is automatically loaded from your account
- All platform features are accessible: PvP Battles, Practice Gym, Clan Wars, and Leaderboards

1.3.5 Troubleshooting

- **Invalid credentials** - Verify email and password are correct; check for typos
- **Email not verified** - Check email inbox for verification link; check spam folder
- **Too many attempts** - Wait 15 minutes before trying again

2 User Roles

CodeClash Arena is designed exclusively for authenticated registered users to ensure fair participation in competitive coding battles. Access to the platform requires account creation, email verification, and a linked Codeforces handle. There are no guest or admin roles in the system - all users have equal access to platform features and compete on a level playing field based on their coding skills and performance.

2.1 Registered User

A registered user is any individual who has successfully completed the sign-up process, verified their email address, and linked a valid Codeforces handle to their account. This is the only role on the platform, and all authenticated users operate with the same base permissions.

2.1.1 Permissions

Registered users have access to the following platform features:

- **Account Management** - Register an account with email and Codeforces handle, log in and log out securely, update profile information and preferences
- **Real-Time 1v1 Coding Battles** - Participate in Local battles (challenge specific friends with custom settings), participate in Global battles (matchmaking with players of similar skill level), compete in time-based coding challenges with live synchronization
- **Clan System** - Create a new clan or join existing clans, participate in clan battles when selected by clan leader, contribute to clan rankings and statistics, manage clan memberships (leaders can approve/reject join requests)
- **Code Submission** - Submit code solutions in multiple programming languages (C++, Java, Python, JavaScript, etc.), receive automated verdicts from Codeforces judge (Accepted, Wrong Answer, Time Limit Exceeded, etc.), test code locally using integrated code editor
- **Statistics and Rankings** - View personal battle history and performance metrics, track XP, level, and ELO-based rating progression, access global and clan leaderboards, view detailed submission verdicts and problem-solving statistics
- **Practice Mode** - Access Practice Gym with thousands of Codeforces problems, filter problems by difficulty, tags, and rating range, receive AI-based logical hints for approach guidance (without revealing full solutions), solve problems without time pressure for skill improvement
- **Social Features** - Send and accept friend requests, view friend profiles and statistics, receive notifications for clan join requests and battle invitations

2.1.2 Restrictions

Registered users are subject to the following restrictions to maintain platform integrity and fair competition:

- **No System Modifications** - Users cannot modify system configurations, settings, or platform code
- **No Ranking Manipulation** - Direct manipulation of ratings, XP, leaderboards, or battle outcomes is strictly prohibited
- **No Private Data Access** - Users cannot access other users' private information, submission code, or system database records
- **No Unauthorized Authentication** - Users cannot share accounts or use multiple accounts to gain unfair advantages
- **Fair Play Requirements** - Code submissions must be original work; plagiarism or copying solutions is not permitted

3 Navigation Guide

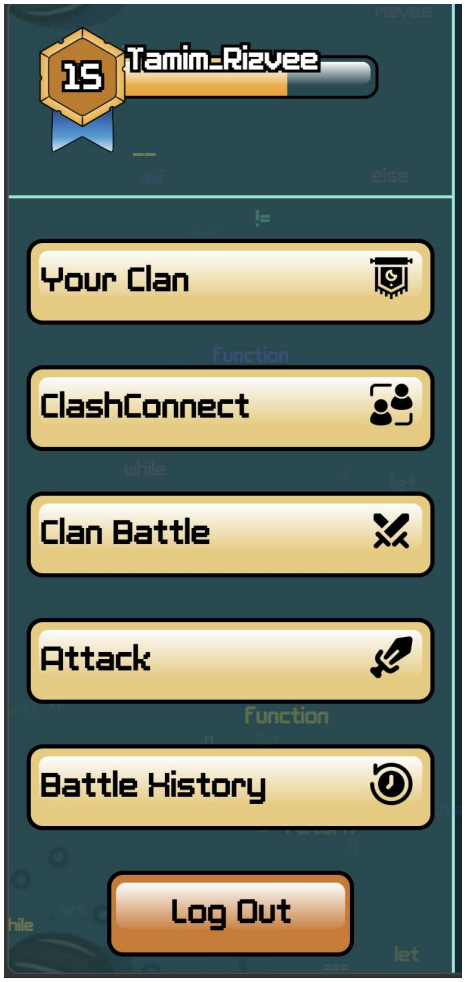
3.1 Menu Structure

After login, the user lands on the main menu dashboard of the game. At the top of the screen, the user's profile information is displayed, showing the player name, the current level, and a progress bar indicating the player's advancement toward the next level.

Below the profile section, the interface presents a vertically arranged menu layout consisting of the following navigation options: **Your Clan**, **ClashConnect**, **Clan Battle**, **Attack**, and **Battle History**. These options are displayed as large, clearly separated buttons to allow quick navigation through the main features of the system.

At the bottom of the menu, there is a **Log Out** button which allows the user to exit the account and return to the login screen. The overall structure provides a simple and organized navigation system for accessing the main functionalities of the application.

> **Note:** The **Card Interface** at the middle of the screen displays what exactly each button does upon hovering them.



(a) Main Menu Buttons

(b) Card Interface

Figure 3.1: Two Interfaces in Landing Page

3.2 Navigation Options

3.2.1 Your Clan

This button launches a floating menu showing the details of current clan the user belongs and details and ratings of the clan members if the current user is in a clan.Otherwise, the floating menu will show two options for finding a clan or creating a clan.

3.2.2 Clash Connect

This button opens a floating menu showing two options. First, it shows a page for finding people and send friend request to them.User can cancel an unaccepted friend request from here. Second, there is a mail page where friend requests sent to user are shown. This menu also shows upcoming clan requests if the user is a leader of a clan.

3.2.3 Clan Battle

This button opens a page where user can participate in (if the leader has chosen current user for battle) or spectate (if the leader has not chosen current user for battle) an ongoing clan battle.

Note: The user must be a member of a clan to avail this feature.

3.2.4 Attack

Launch the PvP battle menu to compete in 1v1 coding challenges. This button opens an overlay menu with multiple battle mode options: Global Battle Mode (matchmaking with players of similar rating), Local Battle Mode (challenge specific friends), and Practice Gym (solo practice without competition). Select your preferred mode to begin competitive coding.

Note: The user must make some friends to participate in the **Local Battle Mode**

3.2.5 Battle History

View your complete battle record and performance statistics. This section displays all your past 1v1 battles and clan battles, showing opponents faced, problems solved, battle outcomes (win/loss), time taken, and rating changes. Use this feature to analyze your performance trends and track your improvement over time.

3.2.6 Log Out

Securely end your current session and return to the homepage. Clicking this button clears your authentication tokens and logs you out of the platform. Always use this option when accessing CodeClash Arena from shared or public computers to protect your account security.

3.3 Buttons and Icons

Button / Icon	Function
Close Button	Closes an overlay menu.
Leave Button	Enables user to leave a clan.
Level Icon	Clicking on the level copies users' unique ID to the clipboard.
Accept Button	Enables user to accept clan or friend requests and local or global battle requests.
Reject Button	Enables user to reject clan or friend requests and local or global battle requests.
Challenge Button	Enables user to send a 1v1 challenge to a friend.
Send Request Button	Enables user to send friend request to an existing user or clan request to an existing clan.
Cancel Request Button	Enables user to cancel friend request to an existing user or clan request to an existing clan.
Enter Button	Enables user to enter 1v1 battle mode or practice gym.
Exit Button	Enables user to exit from 1v1 battle mode or practice gym.
Submit Button	Enables user to submit a problem to Codeforces.
Search Button	Start the search according for caln or a existing user.
View Detail button	Hovering this button shows friends' details.

Table 3.1: Functions of Different Buttons and Icons

4 Error Handling and Messages

4.1 Authentication Errors

Error Message	Possible Cause	Solutions
"Invalid login credentials"	The entered email or password is incorrect, or formatting issues like extra spaces or wrong capitalization exist.	• Check email and password • Ensure Caps Lock is off • Remove extra spaces • Use email instead of Codeforces handle • Verify email after signup
"Email not confirmed" or "Error fetching user data"	The account has not been verified via email, preventing successful login.	• Check inbox or spam for verification email • Click confirmation link • Retry login after verification
"Too many login attempts"	Multiple failed login attempts have temporarily locked the account for security reasons.	• Wait 15–30 minutes before retrying
"User already registered" or "Email already exists"	The email is already associated with an existing account.	• Use login page • Reset password if forgotten • Use another email for new account
Field validation error or "Invalid handle"	The Codeforces handle provided is incorrect or does not exist.	• Verify handle on Codeforces profile page • Check spelling and capitalization • Create account if needed

Table 4.1: Common Login and Authentication Errors

4.2 Network and Connectivity Errors

Error Message	Possible Cause	Solutions
"Network error" or "Failed to fetch"	The browser or application could not connect to the server due to internet issues, VPN restrictions, firewall blocks, or cached data problems.	• Check internet connection • Refresh page (F5) • Verify firewall settings • Disable VPN temporarily • Clear browser cache • Try a different browser
"Request timeout" or "Connection timeout"	The server did not respond in time, often caused by slow or unstable internet, high bandwidth usage, or network congestion.	• Ensure stable connection (5+ Mbps for battles) • Close bandwidth-consuming applications • Switch to a wired connection • Try again after a moment

Table 4.2: Common Network and Timeout Errors

4.3 Clan Related Errors

Error Message	Possible Cause	Solutions
"Already in a clan" or "Join request failed"	The user is currently a member of another clan, or the target clan is not accepting new members or has pending requests.	• Leave current clan before joining another • Wait for pending requests to be processed • Verify clan is accepting members and you meet requirements
"Insufficient permissions" or "Only clan leader can start battles"	The user does not have the required leadership permissions to initiate a clan battle.	• Contact clan leader to start battle • Create your own clan for leadership permissions

Table 4.3: Common Clan-related Errors

4.4 Battle Related Errors

Error Message	Possible Cause	Solutions
"Connection lost" or "Battle disconnected"	The connection between the client and server was interrupted, often due to unstable internet or network congestion. Failure to reconnect may result in automatic forfeit.	• Use wired Ethernet connection • Ensure stable internet before battles • Close unnecessary applications • Avoid battles during network instability
"Submission failed" or "Error submitting code"	The code submission could not reach the server or was rejected due to incorrect configuration, network issues, or account problems.	• Check internet connection • Verify Codeforces handle is correctly linked • Ensure Codeforces account is in good standing • Retry after a few seconds • Verify correct programming language is selected

Table 4.4: Common Battle and Submission Errors

Note: Users must not open any screen recorder, snipping tool, or use third-party applications during code submission. Doing so may prevent the submission from being processed, and the problem may not be fetched and displayed. Codeforces has increased its security to ensure fairness and transparency in competitive programming.